RENEWABLE TODAY.
CLEANER TOMORROW.

ETHANOL

From Fermentation to Fuel

A SIMPLE MOLECULE. A POWERFUL IMPACT.

Renewable Resource | Versatile Applications | Cleaner Energy Future

Fuels People. Powers Progress.

1 COMPOUND IDENTITY

Common name: Ethanol
IUPAC name: Ethanol
Molecular formula: C_2H_6O
Molar mass: 46.07 g/mol
CAS number: 64-17-5

A simple alcohol with big potential.

2 STRUCTURE

$H_3C - CH_2 - OH$

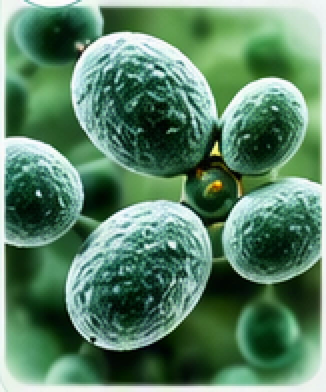
Functional group:
Hydroxyl (−OH)

- Classified as an alcohol
- Polar molecule
- Capable of hydrogen bonding

3 IMPORTANT PROPERTIES

Appearance:	Odor:	Solubility:	Boiling point:	Melting point:
Colorless liquid	Characteristic (alcoholic)	Miscible with water (in all ratios)	78.37 °C (at 1 atm)	−114.1 °C

4 WHAT IS AMAZING ABOUT IT?



Ethanol can be produced naturally by microorganisms such as yeast through fermentation of sugars. The same molecule found in alcoholic beverages can also be used as a renewable fuel, helping reduce dependency on fossil fuels and lower greenhouse gas emissions.

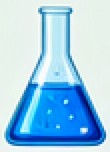
"Tiny microbes create a molecule that moves the world."

5 USES AND VALUE

Alcoholic beverages

Fuel and fuel additives (e.g., E10, E85)

Hand sanitizers


Solvent in industrial and laboratory processes

Chemical intermediate (e.g., for ethyl acetate)

A valuable compound in the food, energy, pharmaceutical, and chemical industries.

6 RISKS AND SAFETY

- Highly flammable liquid and vapor
- Vapors may form explosive mixtures with air
- Keep away from heat, sparks, and open flames
- May cause eye irritation
- Prolonged exposure may cause dizziness or other health effects

Handle and store according to standard laboratory and industrial safety practices.

7 CHEMICAL ENGINEERING CONNECTION

Ethanol production involves fermentation, distillation, and separation processes. Chemical engineers design and optimize:

- Bioreactors for fermentation
- Distillation columns for purification (e.g., azeotropic distillation, dehydration)
- Separation and recovery systems
- Process integration for biofuel and chemical manufacturing
- Waste treatment and sustainability strategies

Engineering solutions for a cleaner, more sustainable future.

8 REFERENCES

PubChem. (n.d.). Ethanol (CID 702). National Center for Biotechnology Information. https://pubchem.ncbi.nlm.nih.gov/compound/Ethanol

National Institute of Standards and Technology. (n.d.). Ethanol. NIST Chemistry WebBook. https://webbook.nist.gov/cgi/cbook.cgi?ID=64-17-5

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